

B.Tech CSE - Data Science

Sample University Course Syllabus for Curriculum AI Testing

Semester: 6

Credits: 4

Course Objectives

Develop strong theoretical knowledge and industry-ready practical skills.

Module 1

Python Programming, NumPy, Pandas, Data Analytics Lifecycle.

Module 2

SQL, PostgreSQL, Data Wrangling, Feature Engineering.

Module 3

Statistics, Probability, Data Visualization, Power BI.

Module 4

Machine Learning, Regression, Classification, Clustering.

Module 5

Big Data, Hadoop, Spark, ETL Pipelines.

Module 6

MLOps, FastAPI, Docker, Cloud Deployment, ChromaDB.

Laboratory

Python, Pandas, NumPy, SQL, Power BI, Apache Spark, Docker.

Course Outcomes

CO1: Analyze datasets.

CO2: Build predictive models.

CO3: Create business insights.

CO4: Deploy data science solutions.